



1/7/2026

ASSIGNMENT DAY 4

GPUs and Cooling Systems.

TASK 1.

- ❖ **Open your own laptop or desktop (or use an online PC hardware simulator if you don't have access to a real system) and identify the GPU model installed. Write down the GPU's brand, model number, and whether it is integrated or dedicated.**

➤ **Identify the GPU Installed in My Laptop :**

 **Laptop :** ASUS TUF Gaming F15 (FX507ZV-LP094W).

<u>Details</u>	<u>Information</u>
GPU Brand	NVIDIA
GPU Model Number	NVIDIA GeForce RTX 4060 Laptop GPU (8 GB GDDR6)
GPU Type	Dedicated (Discrete) Graphics Card

Explanation :

- ✓ My laptop has an NVIDIA GeForce RTX 4060 Laptop GPU, which is a dedicated graphics card. It is designed for gaming, video editing, 3D design, and other graphics-intensive tasks. The laptop also has Intel Iris Xe Graphics, which is an integrated GPU used for everyday tasks and better battery efficiency.

Key Features :

- ✚ Supports real-time ray tracing for realistic graphics.
- ✚ Uses AI-powered DLSS to improve gaming performance and image quality.
- ✚ Includes 8 GB GDDR6 VRAM for modern games and creative applications.

Technical Summary :

- ✓ The NVIDIA GeForce RTX 4060 Laptop GPU delivers excellent graphics performance. It is suitable for gaming, video editing, 3D design, and other high-performance tasks.

TASK 2.

- ❖ **Take a clear photo or screenshot of the cooling system (fan, heatsink, or liquid cooling setup) inside your PC or from a PC hardware simulator. Label the main cooling components and briefly describe their function.**

- **Cooling System Identification :**



Figure : Cooling system showing the fan, heatsink, and other cooling components.

Cooling System Components and Their Functions :

AIO CPU Block

- ✓ Function: It absorbs heat from the CPU and transfers it to the liquid coolant.

AIO Liquid Radiator (360 mm)

- ✓ Function: It removes heat from the liquid and releases it into the air.

System Intake Fans

- ✓ Function: They pull cool air from outside into the PC case.

System Exhaust Fan

- ✓ Function: It pushes hot air out of the PC case.

Passive Heatsinks (VRM/GPU)

- ✓ Function: They absorb and spread heat from the motherboard and GPU to keep them cool.

Technical Summary :

- ✓ The cooling system keeps the CPU, GPU, and other components at a safe temperature. It prevents overheating, improves performance, and helps increase the lifespan of the PC.

TASK 3.

- ❖ **Create a comparison table listing at least 3 differences between air cooling and liquid cooling systems for GPUs, including factors like cost, efficiency, and maintenance.**

- Comparison between Air Cooling And Liquid Cooling For GPUs :

<u>Feature</u>	<u>Air Cooling</u>	<u>Liquid Cooling</u>
Cost	More affordable and budget friendly	More expensive because of additional cooling components
Cooling Efficiency	Provides good cooling for daily use and normal gaming	Provides better cooling for high end gaming and heavy workloads
Maintenance	Easy to install and requires very little maintenance	Requires more maintenance and careful installation

Noise Level	Can become noisy at high fan speeds	Usually quieter because it cools more efficiently
Typical Use	Best for regular gaming PCs and everyday users	Best for high performance gaming PCs and enthusiasts

Technical Summary :

- ✓ Air cooling is an affordable and reliable option for most users. Liquid cooling provides better cooling performance and lower temperatures, making it a good choice for high-end gaming and heavy workloads, but it is more expensive and requires more maintenance.

TASK 4.

❖ **Imagine you are building a gaming PC for streaming IPL matches and playing graphics-heavy games. Choose between air cooling and liquid cooling for your GPU, and justify your choice in 3-4 sentences based on your research.**

➤ **Choosing Between Air Cooling and Liquid Cooling for a Gaming PC :**

- ✓ I would choose air cooling for my GPU. It provides good cooling while streaming IPL matches and playing graphics-heavy games. Air cooling is more affordable, easy to install, and requires very little maintenance. It also offers reliable performance, making it a practical choice for most gaming PCs.

